Model Documentation: Fraud Risk Model fraud_v1 (MDL-004)

Version 1.0 | Fictional document created for the FinGuard AI portfolio project.

1. Model Overview

Model fraud_v1 ranks transactions by relative fraud risk to help analysts prioritise review. It is a logistic regression model with balanced class weights. It is trained on a synthetic dataset for educational use and must not be used for real credit or banking decisions.

2. Training Data

Training used transactions from 9 August to 23 September 2026. Validation used 24 to 30 September. The test period was 1 to 7 October 2026. A time-based split was used to avoid look-ahead leakage.

2.1 Features

Features include log amount, hour of day, a night-time indicator, an international indicator, merchant category, country, channel and device type.

3. Performance

Performance differs sharply by period, and users must consider both results.

3.1 Validation Period

On the validation period, PR-AUC was about 0.05 against a base fraud rate of about 0.76 percent. Normal-period fraud in the training data is only weakly predictable, so the model offers limited ranking power in normal conditions.

3.2 Test Period

On the spike week test period, PR-AUC was about 0.88 and ROC-AUC about 0.98. Among the top 2 percent of alerts, precision was about 0.99, although only about 31 percent of all fraud was captured at that alert budget.

4. Limitations

The risk score is a ranking score, not a calibrated probability, because class weights inflate predicted values. The dataset is synthetic, so results do not predict real-world performance. The model has not been tested for fairness across customer groups. Performance may degrade when fraud patterns shift.

5. Monitoring

Retrain and revalidate when the weekly fraud rate exceeds 2 times the baseline or when validation PR-AUC falls below the documented value.